SiliconGraphics
Computer Systems

Indigo² IMPACT Video™

Indigo²™ Family



Dazzling Digital Video

To satisfy the demanding requirements of post-production and broadcast professionals, Silicon Graphics delivers the 10-bit all-digital Indigo² IMPACT Video™ I/O option board.

With multiple channels of input and output, industry-standard video formats, and real-time video processing, Indigo² IMPACT Video transforms the high-performance Silicon Graphics® Indigo² IMPACT™ workstation into a new category of desk-top tool for the video professional: the first truly capable desktop digital video workstation.

Flexible Video Formats

Send and receive live component video from any serial CCIR-601/SMPTE-259M-compliant device, at the highest quality possible in contemporary video production. Compatible with 525 line (NTSC) and 625 line (PAL) standards, Indigo² IMPACT Video accommodates all major formats of serial digital video I/O.

- 2 channels YUV 4:2:2 (8- or 10-bit)
- 1 channel YUVA 4:2:2:4 (8- or 10-bit)
- 1 channel YUVA 4:4:4:4 (8- or 10-bit)
- 1 channel RP175 (RGBA 8- or 10-bit)
- 2 channels 8-bit arbitrary data

For conversion requirements to and from component or composite analog video, high-quality D/A and A/D solutions from third parties are readily available and affordable. Additional connections provide genlock input, genlock loopthrough, and GPI trigger signals. Utilizing the high-quality genlock, output can be locked to analog house sync or either digital input.

Meeting Fundamental Needs

Building on its broadcast-quality, 10-bit digital video architecture, the distinctive array of features of Indigo² IMPACT Video provides a solid foundation for unlimited applications. Harness the power of Indigo² IMPACT graphics with real-time video and key/alpha output of the 32-bit, double-buffered graphics for broadcast applications. For post-production professionals, Indigo² IMPACT Video can capture and play back uncompressed 10-bit video to and from main system memory.

Expanding Horizons

Combine Indigo² IMPACT Video with the Indigo² IMPACT Compression™ option card to capture digital video signals to disk in real time. Serial digital video I/O coupled with low compression ratios creates an unbeatable combination, dramatically reducing storage and network bandwidth requirements, and enabling demanding applications such as spot playback and non-linear video editing.

Enhance Live Video with Graphics

For the first time, advanced graphics processing and real-time digital video I/O coexist in a desktop environment. The advanced design of this option provides real-time video processing for any combination of sources—computer graphics, rendered animation, and live or stored video. The real-time, 8-bit alpha

blender and key generator enable live creation of many fundamental video effects, including:

- Overlays
- Dissolves
- Fades
- Wipes
- Chroma keying
- Luma keying
- Shadow

Optional Color Space Conversion and Video Texture Mapping

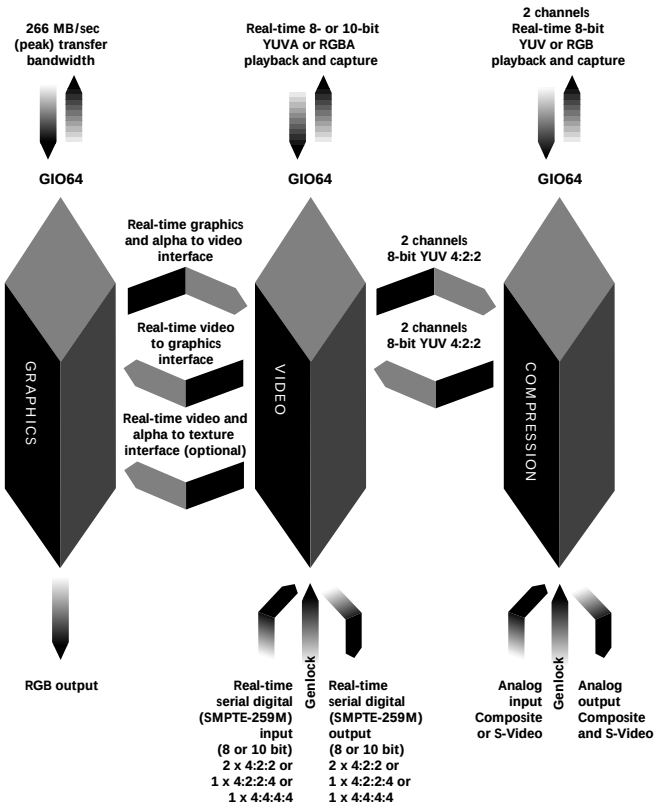
An optional high-quality, 10-bit color space conversion daughterboard gives Indigo^2 IMPACT Video the ability to translate between YUVA and RGBA with high accuracy in real-time. When no alpha channel is supplied, this option also allows you to create alpha channels from superblack YUV. This option also supplies a sophisticated technique for video texture mapping—the ability to use a video signal as a texture map on any 3D object, enabling true 3D digital video effects at refreshingly low costs.

Digital Media Libraries

Providing a reliable foundation for synchronizing audio and video data streams, digital media development libraries (video library, compression library, movie library, and audio library) provide software developers complete control of the video option capabilities. Programmers can access all functions, including:

- Video capture destination and playback sources
- Data formats
- Blending controls
- Genlock controls

ImageVision Library®, Inventor™, and the Developer Magic™ visual development environment complete the software programming toolkit. These interfaces enable programmers to develop the unique applications which help Indigo^2 IMPACT and Indigo^2 IMPACT Video provide the foundation for the digital studio of the twenty-first century.



Indigo^2 IMPACT Video Technical Specifications

System Compatibility
Indigo^2 IMPACT
Single-slot GIO64 card
Internal digital video interface to Indigo^2 IMPACT Video

Input Data Format
Encoding CCIR-601 525/60 and 625/50
Component serial digital (CCIR-601/SMPTE-259M)
±2 active line input acceptance range

Input Signal Characteristics

Signal	Scramble NRZI 270Mb/sec
Connector	Two 75 Ohm BNC
Input Impedance	75 Ohm, terminating, unbalanced
Input Equalization	Capable of equalizing up to 300 meters of Balden 8281 cable at 270Mb/sec

Output Data Format
Encoding CCIR601 525/60 and 625/50
Component serial digital (CCIR-601/SMPTE-259M)
Synchronous and timeable

Output Signal Characteristics

Signal	Scramble NRZI 270Mb/sec
Connector	Two 75 Ohm BNC
Amplitude	800mv±10%
Rise Time	<1ns 20-80%
Overshoot	<10% peak to peak
Clock Jitter	720ps peak to peak 10Hz-10KHz BW

Other Formats Supported
Two channels 8-bit arbitrary data @270Mb/sec
One channel RP-175 (RGBA 8 or 10-bit)

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Silicon Graphics World Wide Web Server
URL: http://www.sgi.com/

Indigo^2 IMPACT Home Page
URL: http://www.sgi.com/Products/Indigo2/IMPACT/

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